Web Crawlers and Robots.txt

How Search Engines Discover and Crawl Web Pages

Yogesh Tanwar 124A1168
Akshata Sonsurkar 124A1157
Samiya Shaikh 124A1147
Shraddha Pawade 225A1203

Introduction to Web Mining

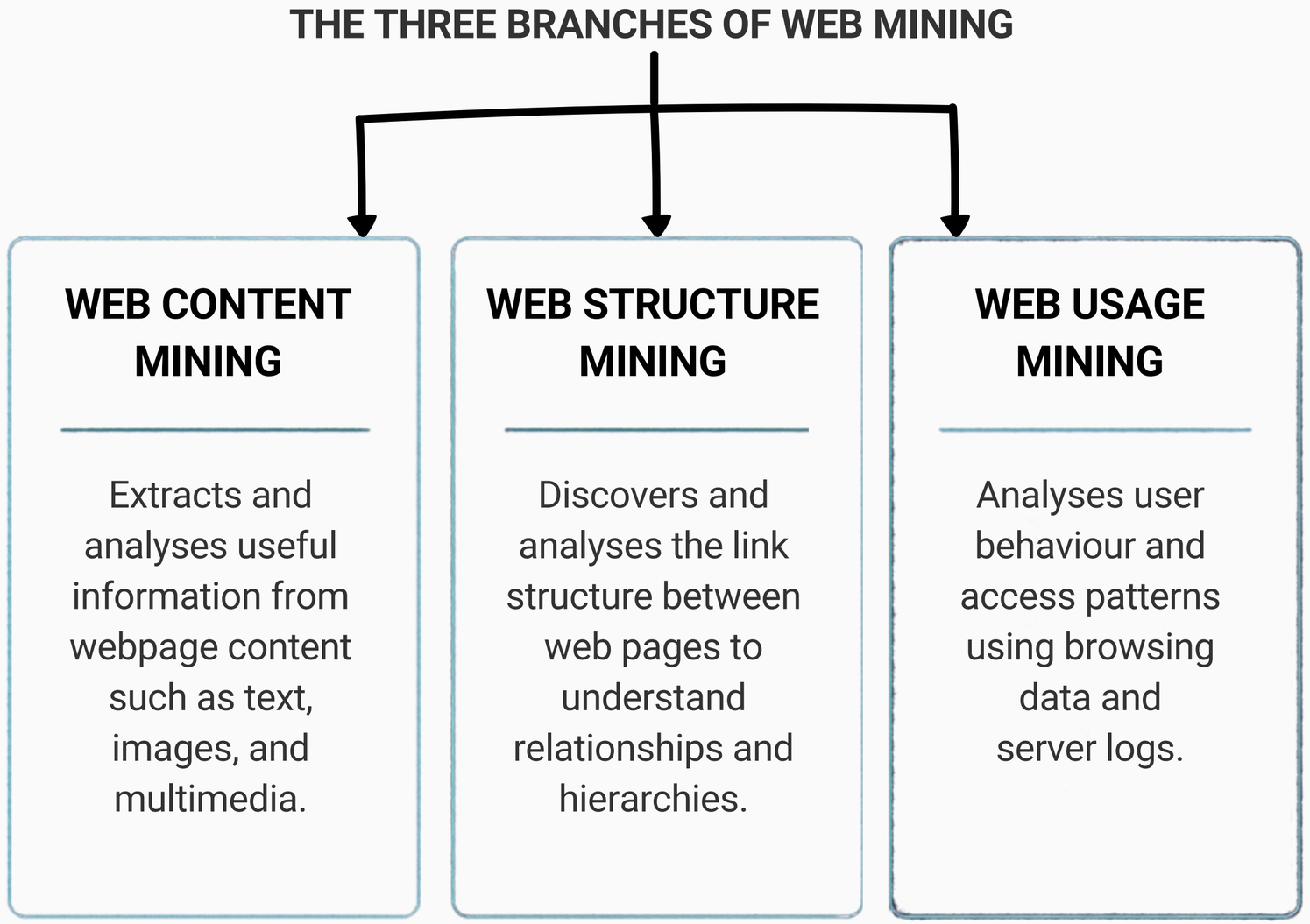
Web mining extracts valuable knowledge from web data through three key approaches.

Web content mining analyses webpage text, images, and other content

Web structure mining studies links between pages

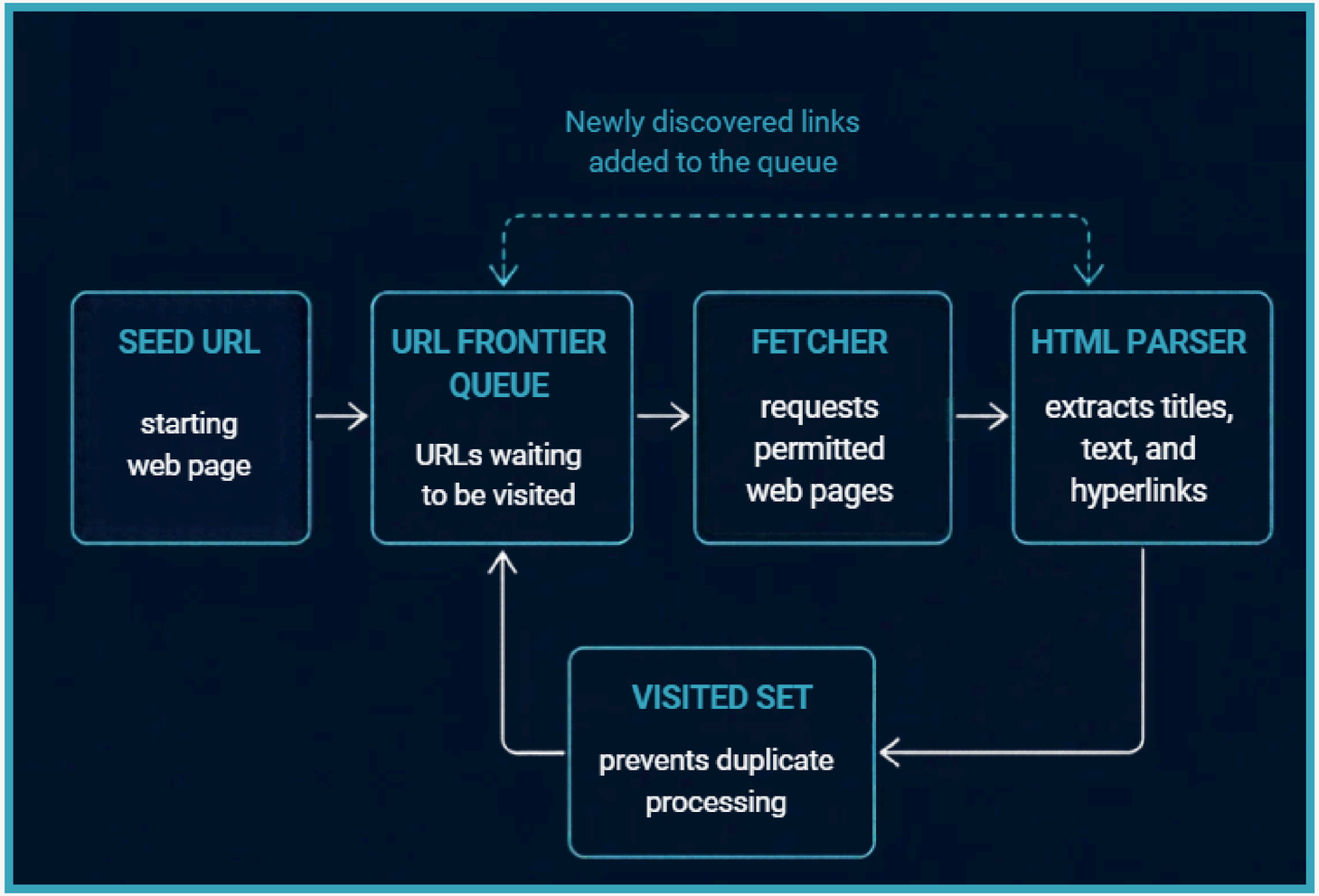
Web usage mining analyses browsing behaviour and access logs

Web crawlers help discover webpages for further processing



Web Crawler Architecture

- Starts from seed URLs and discovers new pages.
- Fetches permitted pages and processes their content.
- Extracts useful data and outgoing links.
- Avoids duplicates and keeps the crawl efficient.
- Newly found links return to the queue for crawling.



BFS Crawling Algorithm

Explore the web level by level using a queue.

Explores pages level by level using a FIFO queue

Visited set prevents repeated visits

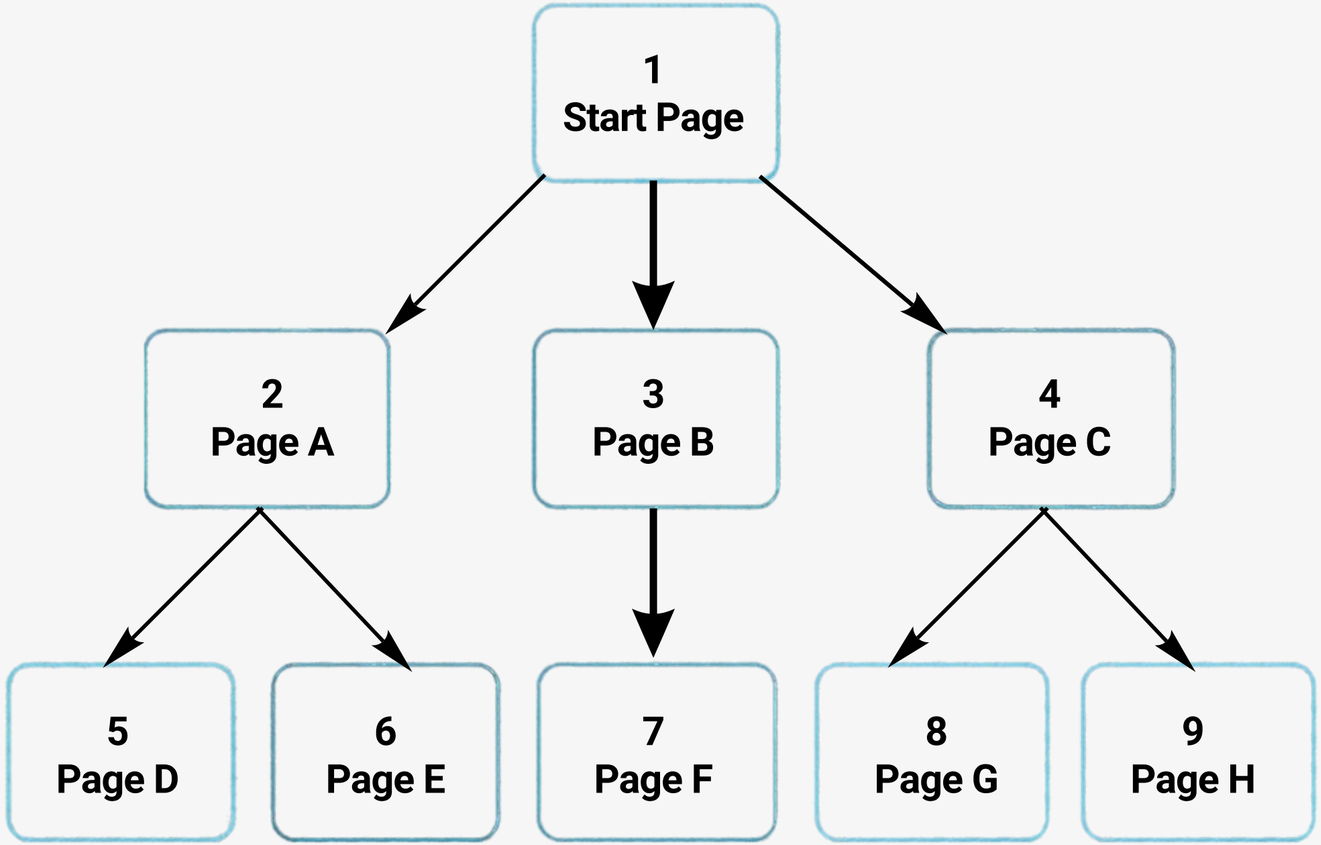
Newly discovered eligible links are enqueued

Crawling stops when the queue is empty or the page limit is reached

PSEUDOCODE

1. Add the seed URL to the queue
2. While queue not empty and page limit not reached:
3. Remove next URL
4. Skip if already processed
5. Check robots.txt rules
6. Fetch page if permitted
7. Extract links and enqueue eligible URLs

VISITATION ORDER (EXAMPLE)



Understanding robots.txt

The robots exclusion protocol tells crewlers what they can and cannot access.

Located at the website root path /robots.txt

User-agent specifies which crawler group

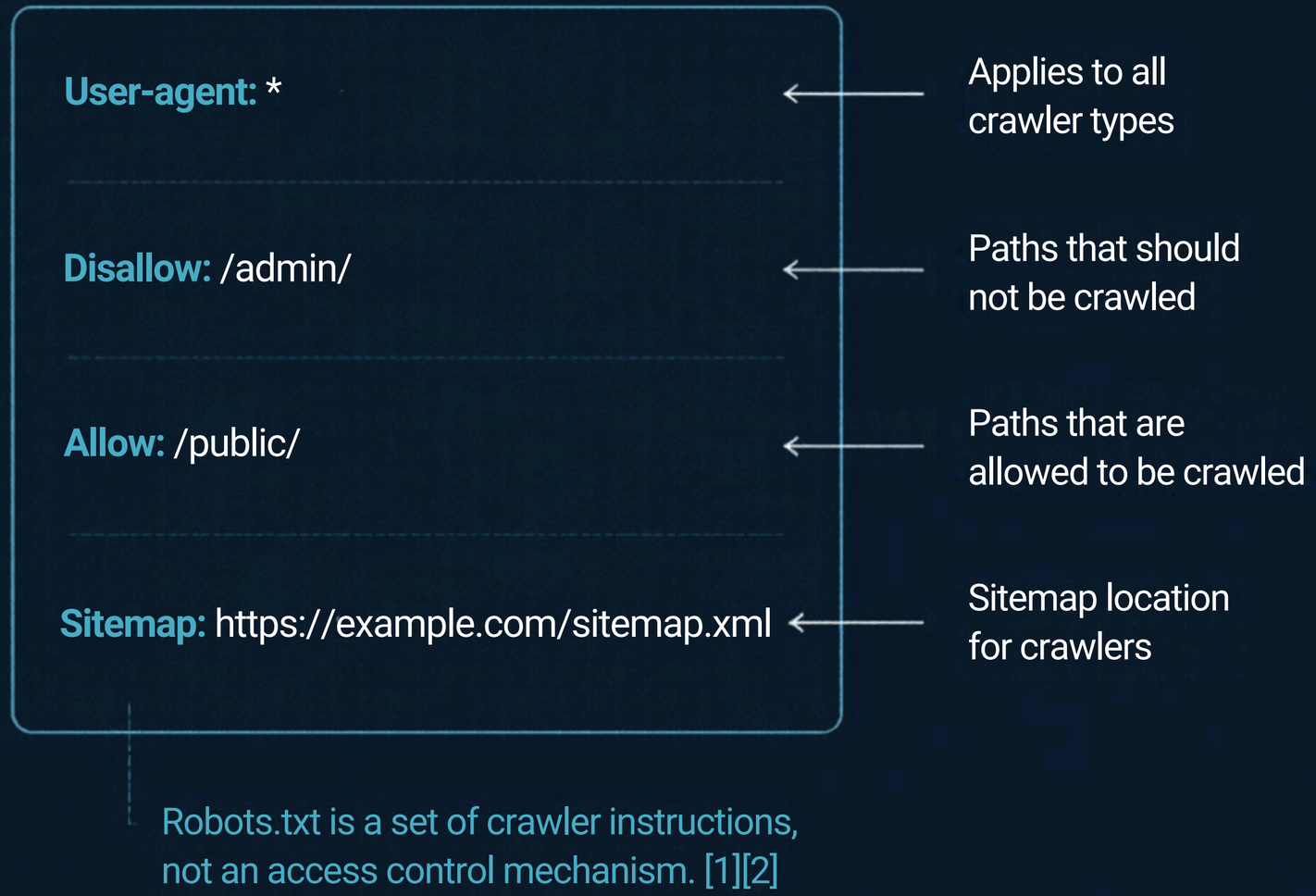
Disallow blocks paths from being crawled

Allow permits specific paths

Sitemap provides the URL of your sitemap

Rules are guidance, not security controls [1][2]

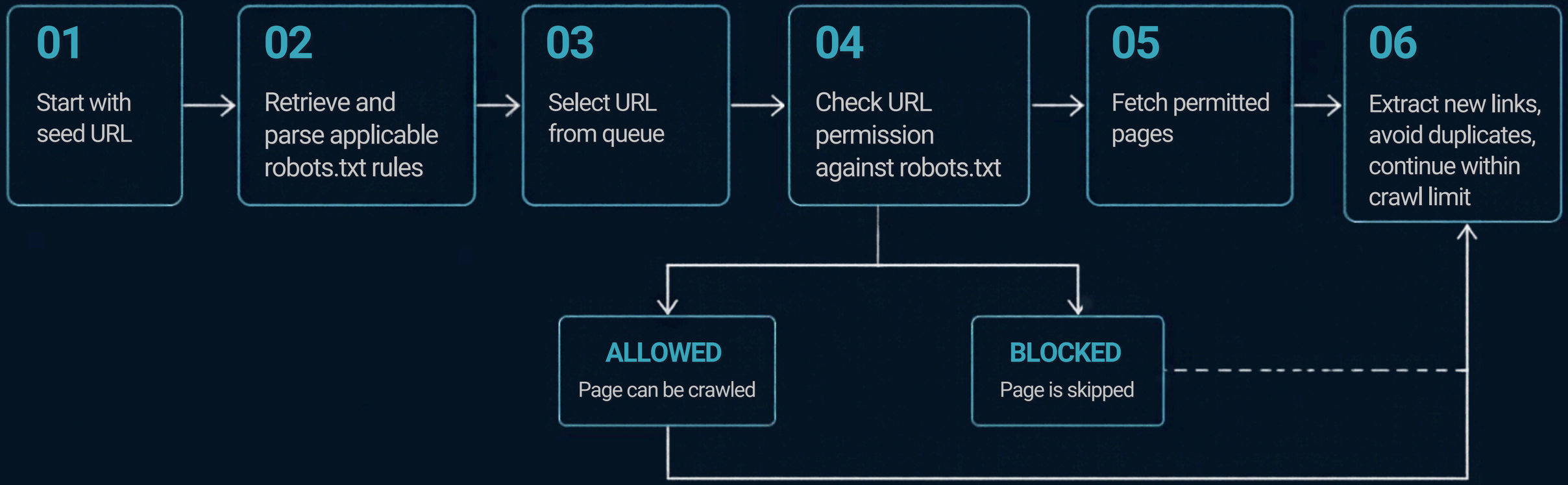
EXAMPLE ROBOTS.TXT (ANNOTATED)



Robots.txt is a set of crawler instructions, not an access control mechanism. [1][2]

How Crawling and robots.txt Work Together

Search engines follow a clear process to discover and respect your site's rules.



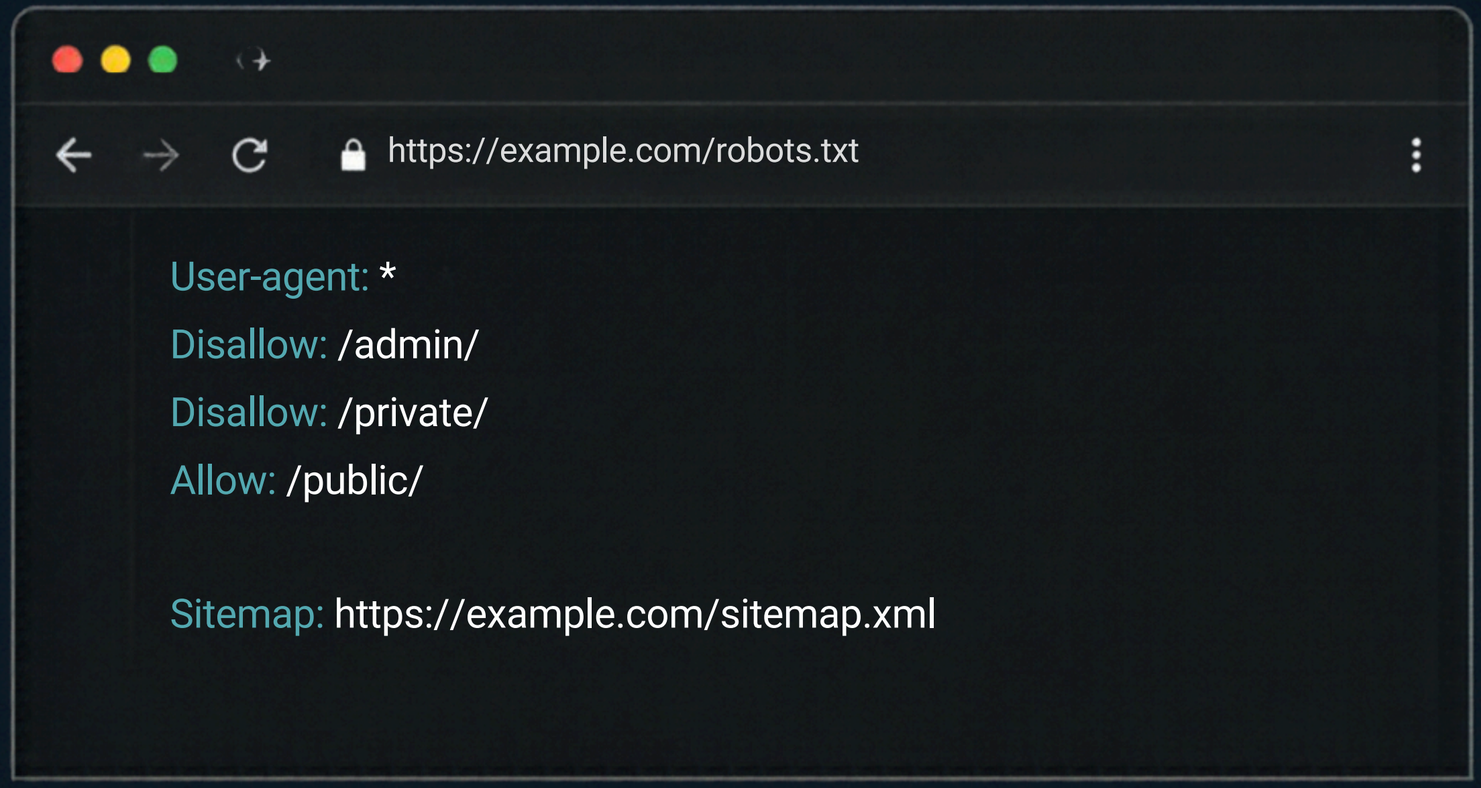
Real-World Case Study

Examining an actual website's robots.txt file

https://example.com

Robots.txt URL:
https://example.com/robots.txt

Date of inspection:
May 12, 2025



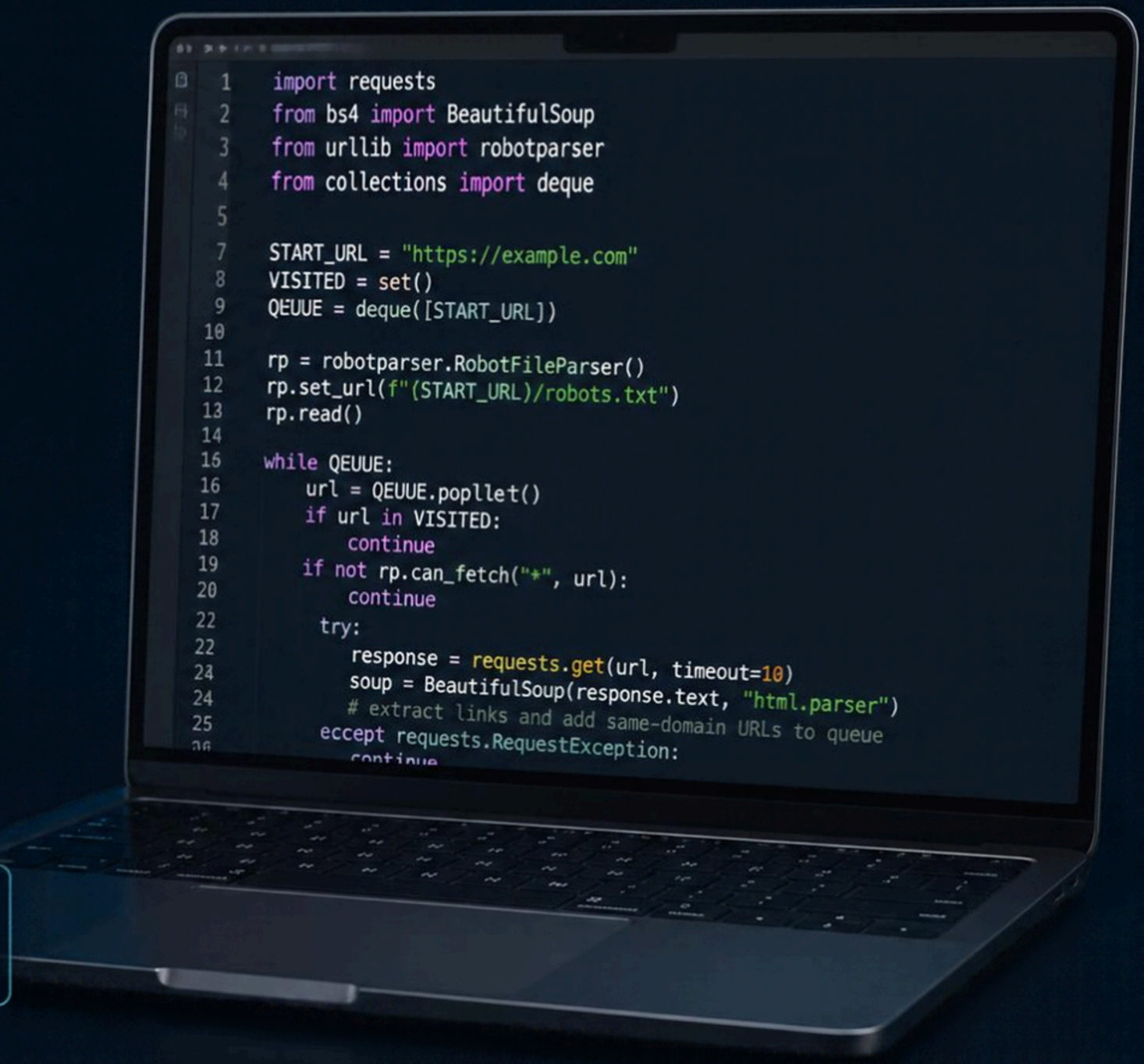
DIRECTIVE (VERBATIM)	EXAMPLE PATH MATCHED		CRAWLER HANDLING (COMPLIANT)
Disallow: /admin/	https://example.com admin/dashboard	→	Must not crawl this path or any URL under /admin/.
Disallow: /private/	https://example.com/private/data/report.pdf	→	Must not crawl this path or any URL under /private/.
Allow: /public/	https://example.com/public/article/123	→	May crawl this path and URLs under /public/.

Sources: https://example.com/robots.txt (inspected May 12, 2025) [1]
RFC 9309: Robots Exclusion Protocol [2]

Proposed Python Implementation

An educational web crawler built with core Python libraries and best practices.

Python with **requests** for HTTP requests
BeautifulSoup for HTML parsing
urllib.robotparser for robots.txt rules
collections.deque for BFS queue
Same domain filtering and duplicate prevention
Timeouts and conservative request spacing
Educational crawler, not a production search engine



OUR PROTOTYPE

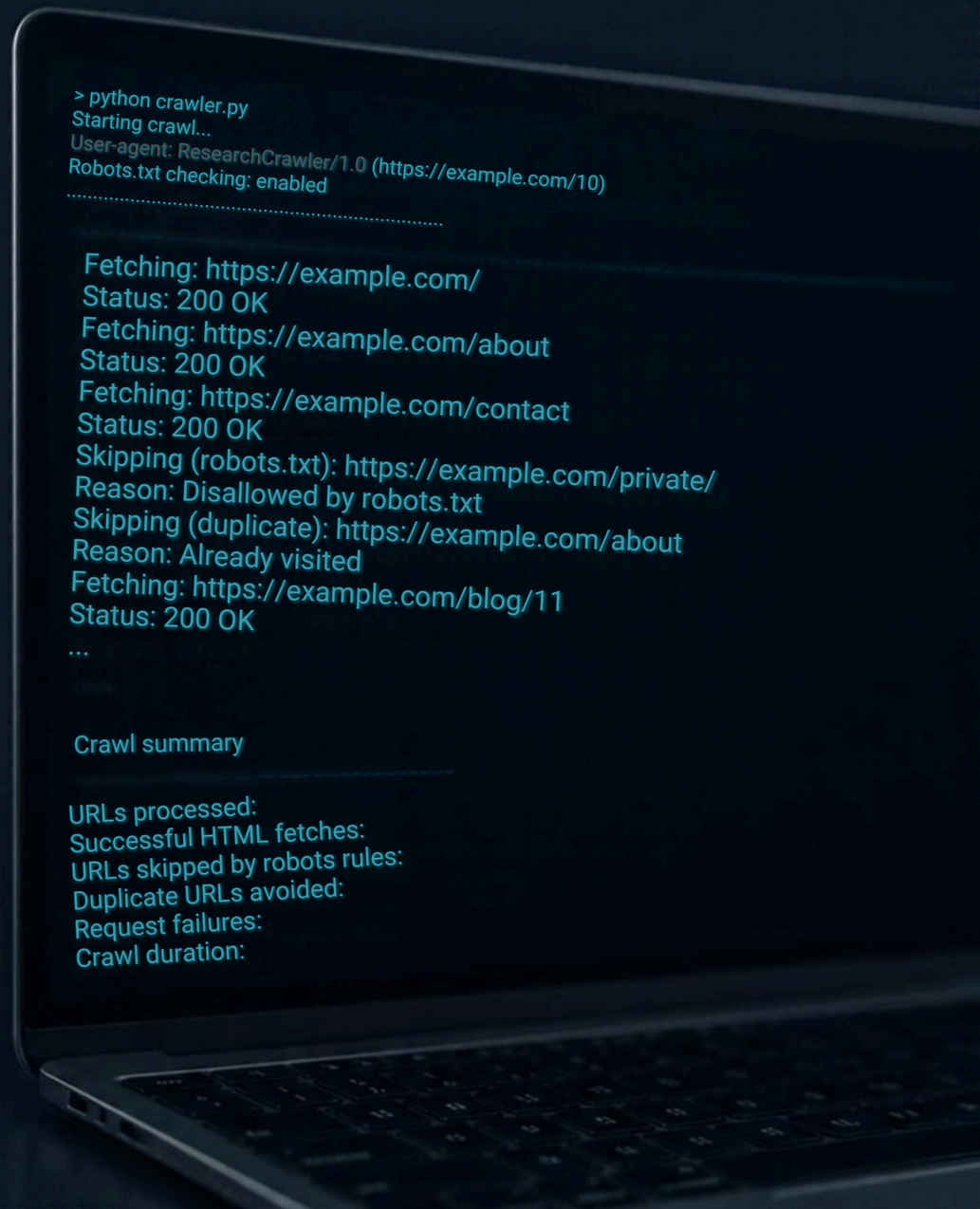
https://github.com/your-username/educational-crawler

Experimental Evaluation

Measuring crawler effectiveness and compliance

Use this results table to record outcomes from your crawler experiments.

Metric	Actual value
URLs processed	
Successful HTML fetches	
URLs skipped by robots rules	
Duplicate URLs avoided	
Request failures	
Crawl duration	



Conclusion and References

Crawlers discover webpages by following hyperlinks

BFS provides a simple traversal strategy

robots.txt communicates crawling rules to compliant crawlers

Queues and visited sets make crawling manageable

Prototype demonstrates rule-aware crawling on limited test website

REFERENCES

[1] IETF, "RFC 9309: Robots Exclusion Protocol," 2022. https://www.rfc-editor.org/rfc/rfc9309.html

[2] Google Search Central, "Introduction to robots.txt." https://developers.google.com/search/docs/crawling-indexing/

[3] Python Documentation, “urllib.robotparser.” https://docspython.org/3/library/urllib robotparser.html

[4] Beautiful Soup Documentation. https://www.crummy.com/software/BeautifulSoup/bs4/doc/

[5] Requests Documentation. https://requests.readtdocs.io/en/latest/

Additional verified IEEE sources to be added after literature check